

Definitions

Procedure

For each term, assemble the following four sections:

- 1. Definition:** Look up the term on Wikipedia. Provide a slightly paraphrased version of their description. Include a reference to Wikipedia and to the sources Wikipedia cites.
- 2. Formula:** Provide the mathematical definition and the full formula associated with it, with all variables defined as in an academic paper. Include a graph visualizing the formula's behavior. Use graphs from Wikipedia when available; cite the source.
- 3. Behavior at Extremes:** Discuss how the system behaves at its extremes using Wikipedia's description. Explain what happens as values approach the limit. Test values numerically (e.g., 10^{-10} , 10^{-100}) to confirm whether computational boundaries are crossed.
- 4. Example:** Provide a real-world example with units. Calculate values approaching the asymptotic limit to demonstrate the limit is never reached. Include a graph visualizing the example. Use graphs from Wikipedia when available; cite the source.

Asymptotic Limit

Definition

In mathematical analysis, asymptotic analysis is a method of describing limiting behavior. If $f(n) = n^2 + 3n$, then as n becomes very large, the term $3n$ becomes insignificant compared to n^2 . The function $f(n)$ is said to be asymptotically equivalent to n^2 as $n \rightarrow \infty$, written $f(n) \sim n^2$.

An asymptote of a curve is a straight line such that the distance between the curve and the line approaches zero as one or both of the x or y coordinates tends to infinity. Vertical asymptotes are vertical lines near which the function grows without bound.

Formula

$$f(x) \sim g(x) \Leftrightarrow \lim_{x \rightarrow a} f(x)/g(x) = 1$$

Vertical asymptote: The line $x = a$ is a vertical asymptote of $y = f(x)$ if:

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \text{ or } \lim_{x \rightarrow a^-} f(x) = \pm\infty$$

Variables: $f(x)$, $g(x)$ = functions being compared; a = the limiting value; $x \rightarrow a^+$ = approach from the right; $x \rightarrow a^-$ = approach from the left.

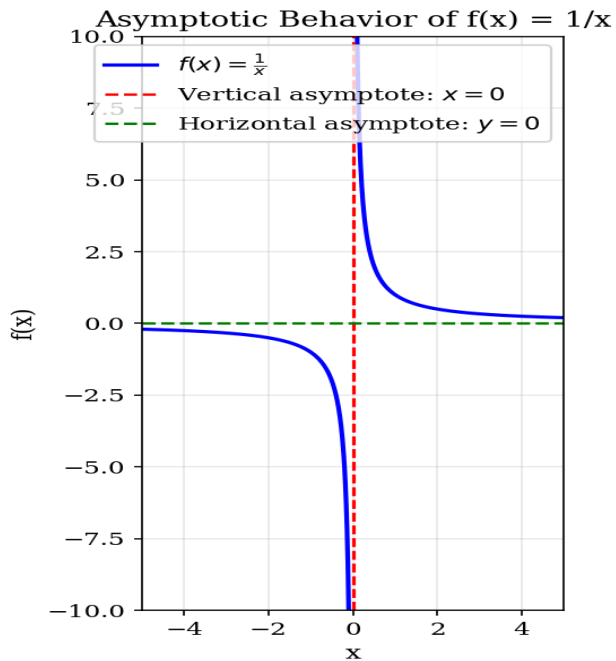


Figure 1: Asymptotic behavior of $f(x) = 1/x$ showing vertical asymptote at $x = 0$ and horizontal asymptote at $y = 0$.

Behavior at Extremes

As values of x become smaller and smaller (approaching zero from the positive side), corresponding values of $y = 1/x$ become larger and larger. The curve extends further upward as it comes closer to the vertical asymptote. No matter how small x becomes, y is never infinite—the limit is never reached.

Numerical test for $f(x) = 1/x$ approaching $x = 0$:

x	$1/x$
10^{-2}	10^2
10^{-3}	10^3
10^{-100}	10^{100}
10^{-1000}	10^{1000}

Example: Newton's Law of Gravitation

The gravitational force between two masses exhibits a $1/r^2$ singularity:

$$F = G \cdot m_1 \cdot m_2 / r^2$$

where $G = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$, $m_1 = 5.972 \times 10^{24} \text{ kg}$ (Earth), $m_2 = 1 \text{ kg}$ (test mass).

$r \text{ (m)}$	$F \text{ (N)}$
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6.371×10^6 (Earth surface)	9.82
1	3.99×10^{14}
10^{-35} (Planck length)	3.99×10^{84}
10^{-100}	3.99×10^{214}

Singularity at $r = 0$: as $r \rightarrow 0$, $F \rightarrow \infty$. The asymptotic limit is never attained.

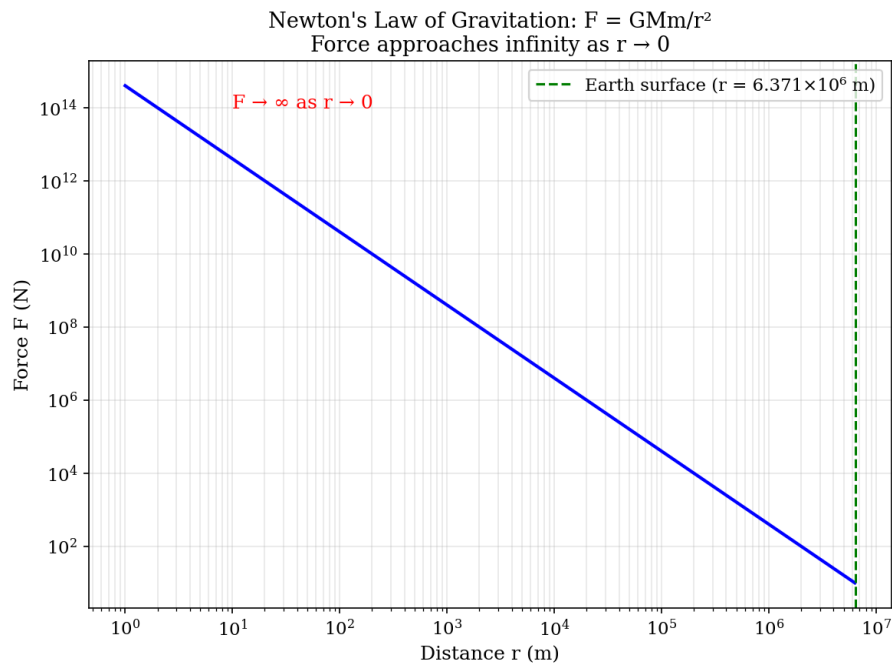


Figure 2: Newton's Law of Gravitation showing force approaching infinity as $r \rightarrow 0$.

References

1. Wikipedia. "Asymptotic analysis." https://en.wikipedia.org/wiki/Asymptotic_analysis
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3. de Bruijn, N.G. (1981). Asymptotic Methods in Analysis, §1.4. Dover Publications.
4. Estrada, R. & Kanwal, R.P. (2002). A Distributional Approach to Asymptotics, §1.2. Birkhäuser.
5. Apostol, T.M. (1967). Calculus, Vol. 1, §4.18. John Wiley & Sons.

Negligibility

Definition

A negligible function $\mu: \mathbb{N} \rightarrow \mathbb{R}$ is a function such that for every positive integer c there exists an integer N^c such that for all $x > N^c$, $|\mu(x)| < 1/x^c$. Equivalently, μ is negligible if for every positive polynomial $\text{poly}(\cdot)$ there exists an integer $N_{\circ} \square_{\gamma}$ such that for all $x > N_{\circ} \square_{\gamma}$, $|\mu(x)| < 1/\text{poly}(x)$.

In measure theory, a negligible set is a set that is small enough to be ignored for some purpose. Finite sets are negligible when studying sequence limits; null sets are negligible when studying integrals.

Formula

$\mu(n)$ is negligible $\Leftrightarrow \forall c > 0, \exists N^c$ such that $\forall x > N^c: |\mu(x)| < 1/x^c$

Variables: $\mu(x)$ = the negligible function; c = any positive integer (polynomial degree); N^c = threshold beyond which bound holds; $1/x^c$ = upper bound that $\mu(x)$ must fall below.

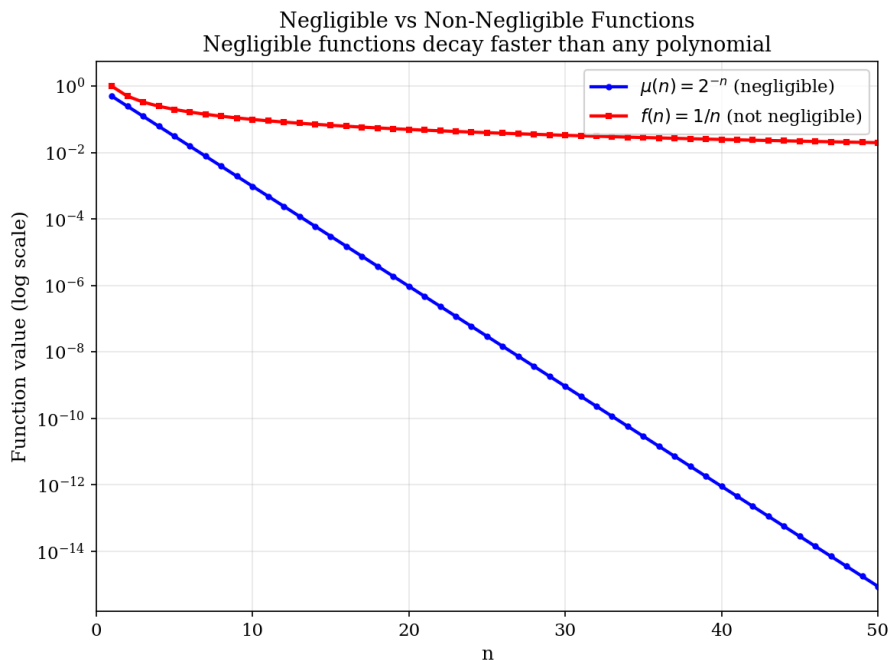


Figure 3: Comparison of negligible function 2^{-n} (blue) versus non-negligible $1/n$ (red).
Negligible functions decay faster than any polynomial.

Behavior at Extremes

Negligible functions approach zero faster than the reciprocal of any polynomial. Exponential decay (e.g., 2^{-n}) is negligible because it decreases faster than any $1/n^c$. Polynomial decay (e.g., $1/n$) is not negligible because it does not decrease faster than all reciprocal polynomials.

Comparison of 2^{-n} (negligible) vs $1/n$ (not negligible):

n	2^{-n}	1/n
10	9.77×10^{-4}	0.1
100	7.89×10^{-31}	0.01
256	8.64×10^{-78}	0.0039

Example: Coulomb's Law

Coulomb's law exhibits the same $1/r^2$ singularity structure. The electrostatic force between two point charges is $F = k_e \cdot q_1 \cdot q_2 / r^2$, where $k_e = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$.

For two electrons ($q = 1.602 \times 10^{-19} \text{ C}$):

r (m)	F (N)
5.29×10^{-11} (Bohr radius)	8.24×10^{-8}
10^{-15} (nuclear scale)	2.31×10^2
10^{-35} (Planck length)	2.31×10^{42}
10^{-100}	2.31×10^{172}

Singularity at $r = 0$: as $r \rightarrow 0$, $F \rightarrow \infty$

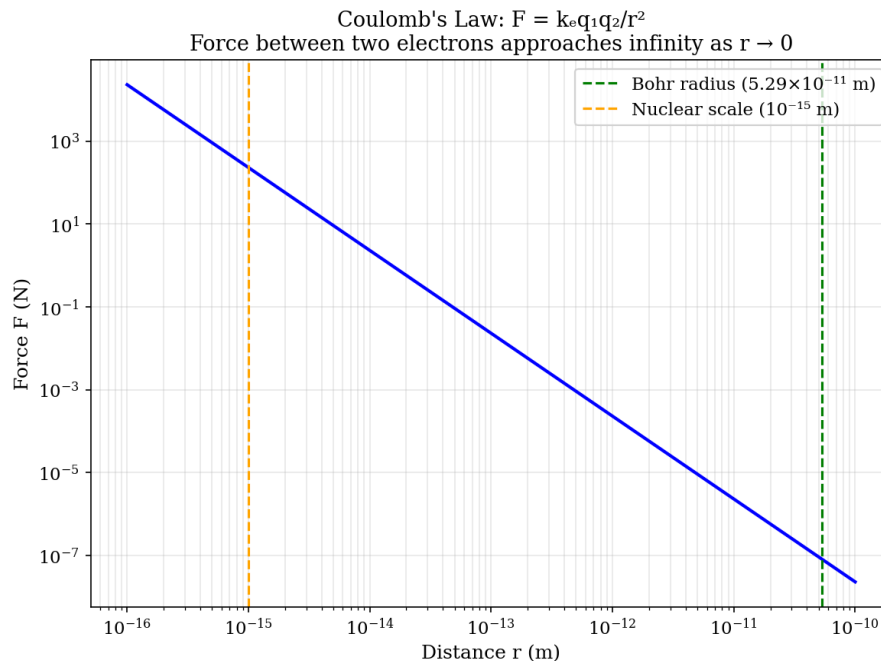


Figure 4: Coulomb's Law showing electrostatic force between two electrons approaching infinity as $r \rightarrow 0$.

References

1. Wikipedia. "Negligible function." https://en.wikipedia.org/wiki/Negligible_function
2. Wikipedia. "Negligible set." https://en.wikipedia.org/wiki/Negligible_set
3. Goldreich, O. (2001). Foundations of Cryptography: Volume 1, Basic Tools. Cambridge University Press.
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5. Bellare, M. (1997). "A Note on Negligible Functions." Journal of Cryptology.

Infinity

Definition

Infinity is something which is boundless, limitless, endless, or larger than any natural number. It is denoted by ∞ , called the infinity symbol. The symbol was introduced by John Wallis in 1655.

In real analysis, the symbol ∞ is used to denote an unbounded limit. The notation $x \rightarrow \infty$ means that x increases without bound, and $x \rightarrow -\infty$ means that x decreases without bound. Infinity is not itself a real number.

The extended real number system is obtained from $\mathbb{R}$ by adding two elements $+\infty$ and $-\infty$ that are respectively greater and lower than every real number. This allows treating potential infinities of infinitely increasing sequences as actual infinities.

Aristotle (350 BC) distinguished potential infinity from actual infinity. At the end of the 19th century, Georg Cantor enlarged the mathematical study of infinity by studying infinite sets, showing they can be of various sizes (cardinalities). The first transfinite cardinal is aleph-null ($\aleph_0$), the cardinality of the natural numbers.

Formula

Limit notation:

$\lim_{(x \rightarrow \infty)} f(x) = L$ means: $\forall \varepsilon > 0, \exists M$ such that $x > M \Rightarrow |f(x) - L| < \varepsilon$

$\lim_{(x \rightarrow a)} f(x) = \infty$ means: $\forall N > 0, \exists \delta > 0$ such that $0 < |x - a| < \delta \Rightarrow f(x) > N$

Extended real arithmetic (where defined):

- $a + \infty = \infty$ for all $a \in \mathbb{R}$
- $a \cdot \infty = \infty$ for $a > 0$; $a \cdot \infty = -\infty$ for $a < 0$
- $a / \infty = 0$ for all $a \in \mathbb{R}$

Indeterminate forms (undefined): $\infty - \infty, 0 \cdot \infty, \infty/\infty, 0/0$

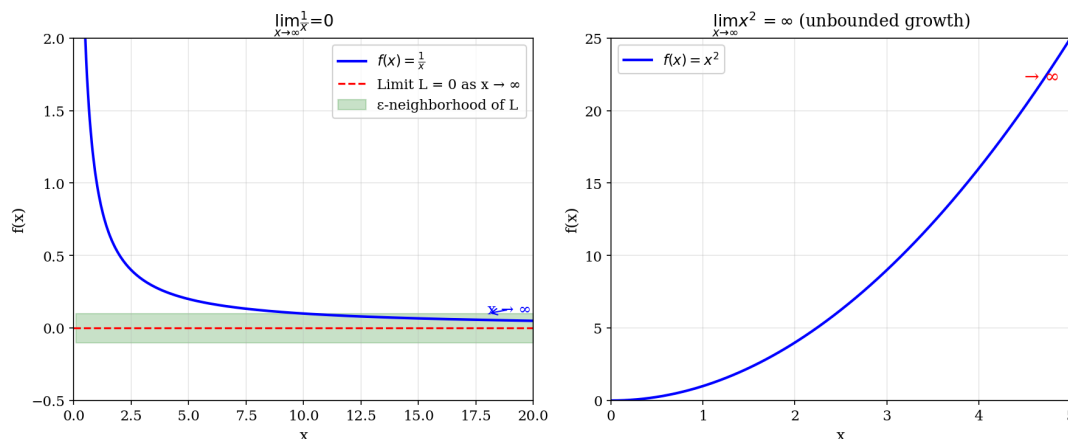


Figure 5: Left: $1/x$ approaching limit $L=0$ as $x \rightarrow \infty$. Right: x^2 growing without bound (limit $= \infty$).

Behavior at Extremes

Infinity is not a destination that can be reached—it is a description of unbounded growth. No matter how large a finite number becomes, it remains finite. The defining property is that for any real number M , there exists a value exceeding M .

Numerical demonstration of unbounded growth:

n	n + 1
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10^{100} (googol)	$10^{100} + 1$
$10^{(10^{100})}$ (googolplex)	$10^{(10^{100})} + 1$
$10^{(10^{(10^{100})})}$	$10^{(10^{(10^{100})})} + 1$

For any n , $n + 1 > n$. The process never terminates. Infinity is never reached.

Example: Harmonic Series Divergence

The harmonic series $H(n) = 1 + 1/2 + 1/3 + \dots + 1/n$ diverges to infinity. Though each term decreases, the sum grows without bound as $n \rightarrow \infty$.

n	$H(n) \approx$
10	2.93
10^3	7.49
10^6	14.39
10^{100}	≈ 230.8
$10^{(10^{100})}$	$\approx 2.30 \times 10^{100}$

Using the asymptotic approximation $H(n) \approx \ln(n) + \gamma$ (where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant), the sum grows logarithmically but without bound. No matter how large $H(n)$ becomes, it remains finite. Infinity is the limit, never attained.

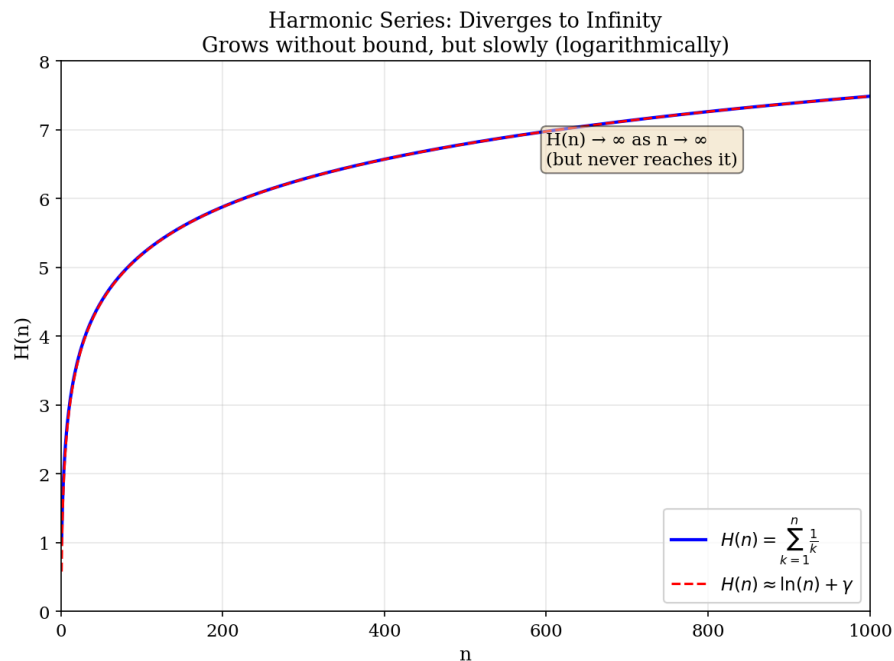


Figure 6: Harmonic series $H(n) = \sum(1/k)$ diverging to infinity. Grows without bound but slowly (logarithmically).

References

1. Wikipedia. "Infinity." <https://en.wikipedia.org/wiki/Infinity>
2. Wikipedia. "Extended real number line." https://en.wikipedia.org/wiki/Extended_real_number_line
3. Allen, D. (2003). "The History of Infinity." Texas A&M Mathematics.
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Figure Attribution

All figures in this document were generated using Python with matplotlib and numpy. The source code is available in `generate_graphs.py`.